

Zhi Wang

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Education

University of California, San Diego
Ph.D. in Data Science

San Diego, CA, USA
Sept. 2025 – Ongoing

University of Science and Technology of China
B.S. in Mathematics, School of Gifted Young

Hefei, China
Sept. 2021 – Jun. 2025

University of California, Berkeley
Visiting Undergraduate Student

Berkeley, CA, USA
Jan. 2023 – Dec. 2023

Research Interests

I am interested in agentic training and inference (e.g. RL algorithms and LM post-training methods), AI for math (e.g. Lean prover training frameworks), multi-agent systems, and learning mechanics. If possible, I would love to learn more about robotics and ML system/infra as well.

Publications

- Eon Lee, Andrés R. Vindas-Meléndez, and **Zhi Wang**¹, *Generalized Snake Posets, Order Polytopes, and Lattice-Point Enumeration*, Discrete Mathematics (2026): 115072.

Ongoing Projects

- Working on writing a Lean prover training framework that supports graph search based iteration, with an emphasis on throughput and flexibility. In parallel research ways to check Lean autoformalization to compliment the training framework in a data flywheel. Working to train a good specialized Lean prover with graph search expert iteration plus other post training methods.
- Working on an online gated controller for CoT degeneracy control in reasoning models, by degeneracy I mean overthinking and looping. This is mostly probing and steering work.
- Working on training a SWE agent via self on-policy distillation, aiming to extend SDPO/OPD to multi-turn setting, and check if a small model can be trained better given this denser signal in an agentic setting.

Research Experience (Undergraduate)

Ehrhart Theory of Order Polytopes
Mentor: Andrés R. Vindas Meléndez, UC Berkeley

Jun. 2023 – Sept. 2024

- Proved closed-form and combinatorial formulas for a family of order polytopes, establishing key base cases and monotonicity results by introducing a novel lattice path coloring method to connect geometric objects with combinatorial enumerations.

¹In mathematics, authors are conventionally listed alphabetically, signifying equal contribution.

Accelerating Grokking via Low-Rank Structures

May. 2024 – Oct. 2024

Mentor: Difan Zou, University of Hong Kong

- Investigated the NTK-to-rich regime transition, leveraging low-rank initialization in MLPs and transformers to accelerate grokking for various tasks.

Simulate Worm Neural activity

Sept. 2024 – Jun. 2025

Mentor: Qian Wen, University of Science and Technology of China

- Trained RNN simulators based on *C. elegans* connectome to reproduce neural activities, carried out ablation study and visualization to guide training principles for neural simulators.

Open Source Contributions

Terminal Bench 2

ICLR 2025

- Implemented two medium level tasks to the benchmark, [PR1](#) and [PR2](#).

Volunteering and Seminars

- Berkeley AI Hackathon 2023 volunteer.
- AI for Mathematics: Formalization and Theorem Proving Seminar. Peking University BICMR, Jan 14th - 27th, 2024.